

ROBERTO CORIA VARGAS

Computational Robotics Engineer | Full Stack Developer

Mérida, Yucatán | +52 984-531-6481 | roberto21coria02@gmail.com | [Portfolio](#) | [LinkedIn](#)

Computational Robotics Engineer with hands-on experience in embedded systems, industrial automation, and full stack web development. I build complete systems — from embedded firmware to APIs and real-time dashboards. Currently working in industry as an electromechanical technician with measurable production efficiency impact. GPA: **9.5 / 10**.

PROFESSIONAL EXPERIENCE

Reyma del Sureste | Electromechanical *Technician*

March 2025 – Present

- Designed and implemented a pressurized-water leak detection system for molds, eliminating disassembly for inspection and reducing pre-inspection time by an average of **3 hours per cycle**.
- Preventive and corrective maintenance on high-speed thermoforming lines (PP cups 5.5 to 16 oz) and industrial mills.
- Technical documentation of internal processes through assembly and disassembly manuals for the mold area.

WINPRO | Systems *Technician (Freelance)*

2020 – 2024

- Hardware and software fault diagnosis and resolution, identifying root causes for effective corrective maintenance.
- Administration and configuration of Linux and Windows operating systems in client environments.

PROJECTS

Personal | Zero Trail — *Electric Vehicle Catalog*

2026

- MERN platform with a database of over **3,000 records** structured by model variants (battery, range, motor). Dynamic filtering and premium-level UI. Final development stage (more information in the portfolio link).

Personal | Mentec Labs — *Desktop optimizer*

2026

- A Windows desktop optimization application built with Electron, React, and TypeScript. Improves operating system performance by up to 25% (more information in the portfolio link).

UPY | Multi-task *robot*

2024

- ROS Publisher/Subscriber architecture in Python and C++; Raspberry Pi 4 as master node coordinating sensors and actuators.
- Trained a custom YOLO model** using a self-captured dataset (local area images) via Roboflow for real-time object detection and tracking with OpenCV.
- SolidWorks CAD design, 3D-printed chassis, remote access via VNC/SSH, dual manual/autonomous control via Tkinter interface.

UPY | Classification *System with Machine Vision*

2023

- 3 Siemens S7-1200 PLCs** interconnected via PROFINET in TIA Portal, managing part sorting by material (metal/plastic) and condition (good/defective).
- COGNEX In-Sight vision system for quality validation; **Mitsubishi RV-4FR robotic arm (6 DOF)** for final part placement.
- Conveyor belt with 5 industrial sensors (fiber optic, inductive, capacitive, reflective, diffuse) and Siemens MicroMaster 420 inverter.

EDUCATION

Universidad Politécnica de Yucatán | *Eng. Computational Robotics Engineering*

Aug 2021 – Apr 2025

- GPA: **9.5 / 10** | State Robotics Tournament I6: Maze Solver, Task Robot, and Battle Robot categories.

English certification: ITEP Plus B2 — December 2024

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML5, CSS3, Java, TypeScript, Google App Script.

Robotics & embedded: ROS, ESP32, Raspberry Pi, Arduino, LoRa, UART, I2C, OpenCV, YOLO/Roboflow

Industrial automation: Siemens S7-1200 PLCs, TIA Portal, PROFINET, Mitsubishi RV-4FR, COGNEX In-Sight

Web development: React, Node.js, Express.js, MongoDB, REST API, MUI, Docker, Git

Tools: SolidWorks, LabVIEW, Linux (Ubuntu), Docker, VNC, SSH